

An Introduction to the Alt-A Sector

July 12, 2007

2nd Edition**Sharad Chaudhary**

212.583.8199

sharad.chaudhary@bankofamerica.com

RMBS Trading Desk Strategy**Ohmsatya Ravi**

212.933.2006

ohmsatya.p.ravi@bankofamerica.com

Qumber Hassan

212.933.3308

qumber.hassan@bankofamerica.com

Vipul Jain

212.933.3309

vipul.p.jain@bankofamerica.com

Sunil Yadav

212.847.6817

sunil.s.yadav@bankofamerica.com

Ankur Mehta

212.933.2950

ankur.mehta@bankofamerica.com

RMBS Trading Desk Modeling**ChunNip Lee**

212.583.8040

chunnip.lee@bankofamerica.com

Marat Rvachev

212.847.6632

marat.rvachev@bankofamerica.com

- ▶ The alt-A sector has matured into the Non-agency MBS market's equivalent of the Specified Pool sector in the Agency MBS market. However, much like the specified pool sector, understanding the nuances of the "convexity story" is crucial. In particular, the diversity of underwriting approaches employed by issuers participating in the sector requires that investors understand how to differentiate between the different alt-A programs. This primer compares and contrasts the different alt-A programs and explains why the "convexity story" holds.
- ▶ The mix of mortgage products in the alt-A sector has changed significantly over the last few years. After playing a relatively minor role in the early years of the alt-A sector, ARMs now represent over 70% of total alt-A issuance and owe their popularity to lowering their affordability barriers through product innovations like Interest-only (IO) and Option ARMs.
- ▶ The key contributing factors behind the attractive convexity characteristics of alt-A deals are the muted refinance responsiveness and higher baseline prepayment speeds of alt-A borrowers relative to borrowers in agency and jumbo pools.
- ▶ The refinancing profile of alt-A borrowers itself results from their weaker credit characteristics from both a credit score and collateral perspective. The weaker credit profile of the alt-A borrower also translates into greater losses on alt-A pools as compared to Jumbo pools.
- ▶ The Alt-A sector is highly leveraged to home price appreciation (HPA) from the standpoint of prepayments and credit. Alt-A borrowers are highly dependent upon strong HPA since it makes it easier to sell an investor property (often financed through an alt-A loan), do a cashout refinance, or refinance into a new mortgage loan with a lower rate. All of these types of voluntary prepayment behavior become more challenging to execute as the housing market cools off. HPA is also important for alt-A loans from a credit standpoint because of the higher leverage and weaker credit profile of the underlying borrowers.
- ▶ The recent downturn in mortgage credit performance has led to market observers making sweeping generalizations about the alt-A and subprime sectors which mask the considerable variation in delinquency rates at the issuer and shelf levels. We analyze the seemingly simple trends in mortgage delinquencies to show the complexity that lies beneath. A close look at alt-A delinquency rates at the issuer and shelf levels shows that there is a significant variation in the credit performance of loans categorized as "alt-A" depending upon who originated these loans.

This document is NOT a research report under U.S. law and is NOT a product of a fixed income research department. This document has been prepared for Qualified Institutional Buyers, sophisticated investors and market professionals only.

To our U.K. clients: this communication has been produced by and for the primary benefit of a trading desk. As such, we do not hold out this piece of investment research (as defined by U.K. law) as being impartial in relation to the activities of this trading desk.

Please see the important conflict disclosures that appear at the end of this report for information concerning the role of trading desk strategists.

I. Introduction

The Alternative-A (“alt-A” for short) sector is a subsector of the private-label (“non-agency”) prime MBS sector. In addition to MBS collateralized by alt-A mortgages, the private-label prime MBS sector also has distinct subsectors that consist of MBS backed by jumbo fixed-rate loans, jumbo hybrid ARM loans, jumbo relocation loans, and other loan types. In most cases, the mortgages collateralizing prime private-label MBS are taken out by borrowers with prime credit¹ who are ineligible for being pooled in an Agency MBS either because their loan size exceeds the conforming loan limit, or because they fall outside the underwriting matrices of the Agencies. In particular, a significant fraction of alt-A borrowers have conforming loan balances but do not conform to the standard Agency underwriting guidelines with respect to LTV, loan documentation, occupancy status, property type, etc.

The two originators behind the inception of the alt-A sector were Indy Mac and Residential Funding Corporation (RFC) through their RAST and RALI private-label MBS shelves respectively. The “alt-A” or “Expanded Criteria Underwriting” programs of both originators were largely focused on providing mortgage credit to borrowers who represented “exceptions” to the standard agency and jumbo underwriting matrices. A simple illustration of the growth of the alt-A subsector and the significant role it currently plays in the non-agency prime MBS sector can be found in Figure 1a which plots overall alt-A issuance and alt-A issuance as a fraction of prime non-agency issuance over the past 12 years. The jump in the relative importance of the sector since 2001 can be attributed to the introduction of alt-A MBS programs by several large commercial banks. Another gauge of the increasing liquidity of this sector comes from the volume of alt-A collateral, which has grown from an outstanding volume of \$4 billion at the end of 1995 to \$675 billion by the end of 2006.

Another relatively recent change in the alt-A sector has been the proliferation in the number of mortgage product types. After playing a relatively minor role in the early years of the alt-A sector, ARMs now represent over 70% of total alt-A issuance and owe their popularity to lowering affordability barriers through product innovations like Interest-only (IO) and Option ARMs. Figure 1c shows the relative popularity of the different ARM product types (conventional ARMs, Hybrid ARMs and Option ARMs) within the alt-A sector. Hybrid ARMs represent 50% of the total alt-A ARM universe and, within hybrid ARMs, 5/1s are by far the most popular product.

Side by side with the growth in the alt-A market has been a proliferation in the number of originators participating in the market by securitizing their alt-A production through various shelves. Some shelves issue securities backed by loans created by a specific originator (**issuer shelves**), while others are operated by MBS dealers who pool together collateral from a variety of different originators (**dealer or conduit shelves**). The issuance history of some key alt-A programs can be seen in Figures 2a and 2b.

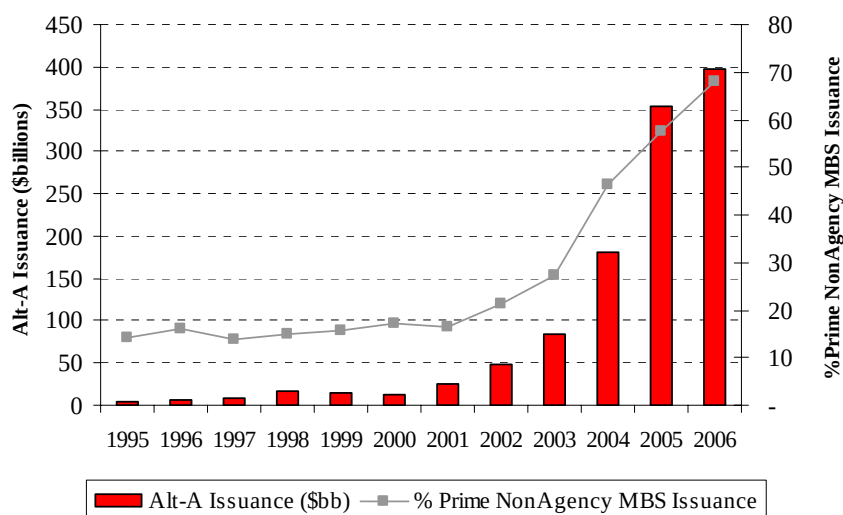
The growth in the number of alt-A issuers poses some challenges for investors. While small differences in underwriting guidelines among originators usually have only a relatively minor impact on the behavior of their conforming or jumbo prime collateral because of the relative uniformity of underwriting practices in these sectors, the

¹ Typically defined to be a borrower with a FICO score over 620.

somewhat subjective nature of what constitutes an alt-A borrower means that in practice there can be substantial differences in alt-A underwriting practices across issuers. Another way to say this is to note that while it is pretty easy to define what is “inside the box,” trying to describe what is “outside the box” is more difficult because it potentially covers a much broader area.

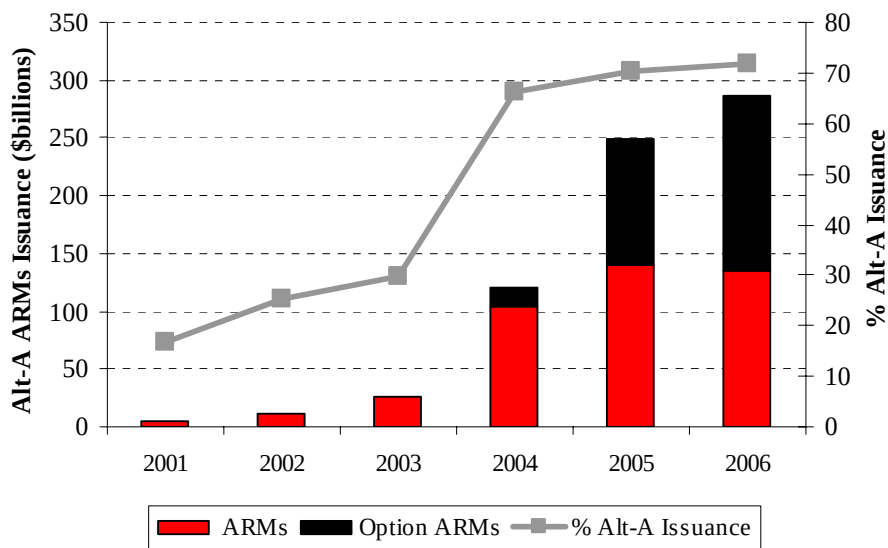
Fortunately, the pandemic of confusion regarding the alt-A sector has created some exceptional opportunities for investors. Many securities backed by alt-A collateral offer very high yields relative to their convexity and credit risk. However, in order to identify the most attractive investment opportunities, it is essential to understand the difference between alt-A and other loan types and how the particular characteristics of alt-A loans vary among originators. The primary goal of this report is to map out these differences and variations and explore how they determine the prepayment and credit characteristics of alt-A pools. The emphasis in this report is thus on providing investors with a feel for the “fundamentals” of the alt-A sector. For brevity’s sake, valuation issues have not been discussed.

Figure 1a: Issuance Trends in the Alt-A Sector (1995-2006)



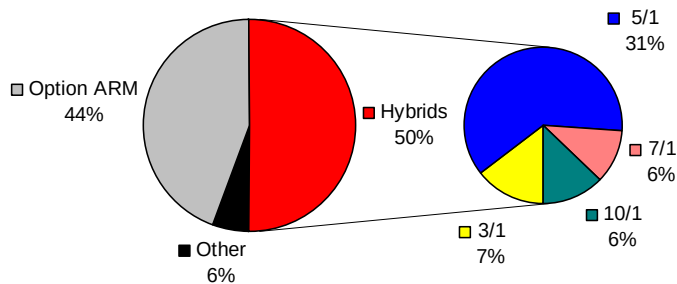
Source: Loan Performance, Banc of America Securities

Figure 1b: Issuance Trends in the Alt-A ARM Sector (2001-2006)



Source: Loan Performance, Banc of America Securities

Figure 1c: Total Outstanding Alt-A ARM MBS by Product Type (as of April 2007)



Source: Loan Performance, Banc of America Securities

Figure 2a: 30-Year Fixed-Rate Issuance Volume for Selected Alt-A Shelves (\$bb)

		Bank Of America	Countrywide	IMPAC	Indy Mac	RFC	Washington
	Total	BOAA	CWALT	IMM/IMSA	RAST	RALI	Mutual WMALT
1996	3.4	0.0	0.0	0.0	2.4	1.0	0.0
1997	5.4	0.0	0.2	0.0	2.5	2.7	0.0
1998	13.4	0.0	1.1	0.0	7.2	5.1	0.0
1999	5.9	0.0	0.8	0.0	1.5	3.7	0.0
2000	6.6	0.0	0.3	1.2	2.1	3.0	0.0
2001	9.9	0.0	4.0	1.5	0.0	4.3	0.0
2002	16.5	0.0	7.6	0.5	2.7	5.7	0.0
2003	24.1	3.9	8.7	0.9	4.0	6.6	0.0
2004	27.0	3.1	16.5	1.6	2.2	3.6	0.0
2005	60.3	4.1	36.6	1.1	6.7	5.7	6.1
2006	55.4	3.1	28.3	1.6	7.2	12.2	2.9

Source: Loan Performance, Banc of America Securities

Figure 2b: ARM Issuance Volume for Selected Alt-A Shelves (\$bb)

		Bank Of America	Bear Stearns	Countrywide	Indy Mac	RFC	Lehman
	Total	BAFC	BALTA	CWALT	INDX	RALI	SARM
2002	0.6	0.0	0.6	0.0	0.0	0.0	0.0
2003	3.5	0.0	3.3	0.0	0.0	0.2	0.0
2004	40.4	1.1	10.2	5.6	3.2	2.2	18.1
2005	53.6	4.2	16.3	3.3	11.2	6.1	12.6
2006	58.6	4.5	17.0	6.9	13.2	5.5	11.4

Source: Loan Performance, Banc of America Securities

II. What is Alt-A Collateral?

Alt-A collateral includes loans from borrowers who have prime A-quality credit but who, for one reason or another, are unwilling or unable to meet all the criteria to qualify for a conventional or jumbo prime loan. Instead, they borrow under the expanded underwriting guidelines of lenders with alt-A programs. Possible reasons why a homeowner may borrow under an alt-A program include:

- Inadequate documentation of assets or income to qualify for the conventional or jumbo prime programs;
- A blemished credit history that disqualifies the borrower from obtaining a conventional or jumbo prime loan in spite of an adequate FICO score;
- The loan-to-value (LTV) ratio for the property or occupancy type falls outside of standard agency underwriting guidelines;
- A loan-to-value ratio over 80% without primary mortgage insurance.

A rigorous definition of what constitutes an alt-A loan is not really practical given the multiple factors that are taken into account in the underwriting process and the idiosyncratic variation in underwriting guidelines from program to program. Matters are further complicated by the fact that in certain cases it is more economic for an originator to securitize a conforming balance loan through the alt-A sector because of the high guarantee fee charged on these loans by the agencies. However, we do attempt to give investors a flavor for some of the general trends one can see in alt-A underwriting in the following section.

III. Collateral Characteristics of Alt-A Programs

As noted above, considerable differences exist across various issuers' alt-A programs, which suggest the need to summarize the types of loan characteristics an investor might see in an alt-A deal. Impatient readers can skip over the detailed examination of the loan attributes discussed below and proceed to Figure 14, which collects all the individual tables. Our analysis is focused on data from originator shelves. Conduit shelves are harder to analyze because they aggregate collateral across different originators, each of whom typically have their own underwriting practices.² Most of the following analysis of the collateral attributes of alt-A programs is focused on 30-year loans, but Figure 15 does summarize the characteristics of 15-year alt-A loans across various alt-A shelves. We also briefly compare and contrast the collateral characteristics of alt-A hybrid ARMs with those of the fixed rate loans. A summary of the characteristics of hybrid ARM alt-A loans across various alt-A shelves is presented in Figure 16.

Documentation

One of the most common reasons for a borrower choosing an alt-A loan is that he or she prefers to submit less documentation in support for the loan application than would be required for a conventional or jumbo prime loan. Lenders categorize such borrowers into broad groups such as "No Income/No Asset", "Stated Income" and "No Ratio". For each of these classes of borrowers, they typically establish maximum loan size and rates based on borrower credit scores and property values (implicitly, debt-to-value ratios). As indicated by Figure 3, the average aggregate percentage of "Full Documentation" borrowers is only 35% for 30-year alt-A borrowers. The typical jumbo 30-year deal has a "Full Documentation" percentage of over 65%.

Figure 3: Documentation Type for 30-year alt-A Collateral

		Bank Of America BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI	Washington Mutual WMALT
Average							
%Full Doc	35%	33%	37%	15%	32%	41%	23%

Source: Loan Performance, Banc of America Securities

Credit Scores

Although an imperfect credit history is probably the second most common reason for a borrower taking out an alt-A loan, the vast majority of alt-A borrowers have credit scores that are consistent with prime borrowers. The average FICO score for the six alt-A issuers profiled in our study was 713, compared to an average FICO score of 730 for the jumbo sector. Figure 4 shows that 22% of all alt-A borrowers have FICO scores below 675 and 10% have FICO scores below 650.³

² The most active conduit alt-A shelves are BSABS (Bear Stearns) and SASC (Lehman). The aggregations for Countrywide's CWALT shelf do include loans from their dealer shelf, which are issued as "J" deals.

³ The large number of missing scores for the CWALT shelf results from the absence of loan level credit scores on Countrywide's 2001 and 2002 deals.

Figure 4: Credit Score Distributions for 30-year Alt-A Collateral

	Average	Bank Of America BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI	Washington Mutual WMALT
Mean	713	733	712	690	707	717	719
% <650	10%	3%	11%	24%	12%	6%	6%
% >=650 and <675	12%	5%	13%	17%	15%	10%	12%
% >=675 and <700	17%	15%	16%	17%	16%	21%	18%
% >=700 and <725	18%	19%	17%	14%	15%	20%	19%
% >=725 and <750	15%	19%	14%	11%	13%	18%	16%
% >=750	25%	38%	25%	15%	21%	25%	28%
% N/A	3%	0%	3%	0%	8%	1%	0%

Source: Loan Performance, Banc of America Securities

Occupancy

On the whole, the proportion of investor-property loans is much higher for alt-A collateral than for either conforming or jumbo prime collateral. Investor properties account for less than 1% of the average jumbo deal and approximately 9% of the average Fannie Mae pool.

Figure 5: Occupancy Type for 30-year Alt-A Collateral

	Average	Bank Of America BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI	Washington Mutual WMALT
%Investor	16%	47%	10%	14%	12%	24%	15%

Source: Loan Performance, Banc of America Securities

Loan Size

Alt-A deals can include conforming balance loans. As a result, the average loan size for alt-A deals tends to be much lower than that of jumbo prime deals, which contributes to the superior convexity of the alt-A sector. The average loan size for 30-year alt-A collateral from the top six issuers in 2006 is about \$254,000. Approximately 66% of 2006 alt-A production had conforming loan sizes as shown in Figure 6.

Figure 6: Loan Sizes for 2006 Vintage 30-year Alt-A Collateral

	Average	Bank Of America BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI	Washington Mutual WMALT
Average (\$000)	254	190	258	259	362	227	250
%Conforming	66%	77%	72%	67%	34%	69%	60%

Source: Loan Performance, Banc of America Securities

Original Loan-To-Value

On average, loan-to-value (LTV) ratios for alt-A collateral are slightly higher than those of agency or jumbo prime collateral. The average loan-to-value ratio for the top six alt-A issuers is 70%. Notice the relatively high proportion of loans with LTVs above 90% in Figure 7. High LTV loans are fairly rare in both Fannie pools and jumbo deals—a snapshot from our databases indicates that around 1% of the loans from these sectors had LTVs that exceed 95%. Approximately 4% of all 30-year fixed alt-A loans fall under this category.

Figure 7: Original Loan-to-Value Distributions for 30-year Alt-A Collateral

	Bank Of America					Washington	
	Average	BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI	Mutual WMALT
Mean	70	67	69	74	68	73	69
% <70	28%	32%	29%	22%	34%	19%	30%
% >=70 and < 80	26%	24%	25%	21%	30%	24%	28%
% >=80 and < 90	36%	34%	36%	33%	29%	42%	38%
% >= 90	11%	9%	10%	25%	7%	14%	3%

Source: Loan Performance, Banc of America Securities LLC.

Loan Purpose

Given the high refinance environment of the past few years, it comes as no surprise that only 47% of all alt-A mortgages are for a home purchase. Of those borrowers that refinance an existing mortgage, approximately 2/3rd do so to take out equity.

Figure 8: Loan Purpose Distributions for 2005 Issuance 30-year Alt-A Collateral

	Bank Of America					Washington	
	Average	BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI	Mutual WMALT
%Purchase	47%	51%	48%	54%	39%	49%	46%
%Cashout Refinance	34%	30%	34%	34%	36%	32%	39%
%Rate-Term Refinance	19%	19%	17%	12%	25%	18%	15%

Source: Loan Performance, Banc of America Securities LLC.

Prepayment Penalty

Sometimes alt-A programs allow borrowers to buy down the rate in return for taking on a prepay penalty. As one might expect, this has quite a significant impact on prepayment speeds. Although prepayment penalties are relatively rare for 30-year fixed-rate alt-As across the different issuer shelves with the exception of loans securitized under the IMPAC shelves (Figure 9), they are very common in alt-A hybrids as indicated in Figure 16. It is worth noting that while speeds of loans with penalties are depressed during the lock-out period, they tend to spike immediately after the initial period for which the penalty is in effect. The typical prepay penalty term for a 30-year fixed-rate loan is 3-5 years but can be as short as one year for alt-A ARMs.

Figure 9: Prepayment Penalty Distributions for 30-year Alt-A Collateral

	Bank Of America					Washington	
	Average	BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI	Mutual WMALT
%HasPrepayPenalty	9%	2%	4%	61%	19%	4%	14%
%PenaltyUnknown	22%	0%	34%	0%	38%	0%	0%

Source: Loan Performance, Banc of America Securities LLC.

Mortgage Rate or Coupon

Given the diversity of alt-A borrowers and the guidelines used to underwrite these borrowers, one would expect a great deal of variability in mortgage rates, even among loans issued during the same time period. The relative premiums imposed for weaker credit or lack of documentation can vary considerably from originator to originator. Other features such as loan size, prepay penalties and location of the property can also influence the rate dramatically. This can make it difficult to calculate the refinance incentive for alt-A collateral, since it is impossible to know precisely what mortgage rates would be available to the underlying borrowers if they choose to refinance. However, in general, the spread that a particular borrower must pay over the prevailing mortgage rate for a particular property shouldn't change too dramatically, barring any change in the borrower's credit profile or the amount of documentation they are able to provide. This can be taken into account by computing the typical spread at origination (SATO) for each deal issuer, i.e., the spread between the average mortgage rate for the underlying loans and the prevailing conforming mortgage rate that was available during the time the loans were issued. In future months, this spread can be added to the prevailing mortgage rate in order to calculate the "true" refinance incentive for the alt-A collateral. Figure 10 collects average SATOs for the issuers profiled in this study.

Figure 10: Average Spread-at-Origination (bp) for 30-year Alt-A Collateral

	Average	Bank Of America BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI	Washington Mutual WMALT
1996	99				101	97	
1997	77		49		71	85	
1998	90		58		90	97	
1999	95		52		118	95	
2000	120		61	191	122	96	
2001	103		92	176	59	88	
2002	71		74	160	48	65	
2003	52	39	48	93	45	62	
2004	55	42	52	113	45	59	
2005	44	40	37	96	66	73	48
2006	59	44	44	109	59	83	64

Source: Loan Performance, Banc of America Securities

Geographic Distribution

Given the somewhat larger balances carried by private-label alt-A borrowers and higher housing costs in the California markets, it is not surprising that they tend to have California concentrations on the high side. As one might expect, California concentrations in alt-A pools lie in between those on agency pools (California concentrations average 17%) and jumbo 30-year pools (California concentrations average 46%).

Figure 11: California Concentrations for 30-year Alt-A Collateral

	Average	Bank Of America BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI	Washington Mutual WMALT
%California	32%	32%	29%	40%	45%	26%	32%

Source: Loan Performance, Banc of America Securities

Seconds

High home prices have resulted in an increasing percentage of loans with seconds in the recent years. Approximately 39% of the alt-A 30-year loans originated during 2006 had seconds (as compared to 27% in jumbo 30-years). Recently, the percentage of seconds in a pool has turned out to be a very important driver of its credit performance. For instance, 2006 origination alt-A 30-year loans with seconds and twelve months of seasoning have 90+ delinquency rates that are 290bps greater than loans of comparable seasoning but without seconds.

Figure 12: Percentage of Loans with Seconds in the 30-year Alt-A Sector

Origination Year	% Seconds
2000	0
2001	2
2002	2
2003	8
2004	17
2005	28
2006	39

Source: Loan Performance, Banc of America Securities

Distinguishing Characteristics of Alt-A Hybrid ARM Collateral

Given that affordability was the primary driving force behind the recent growth of alt-A hybrid ARM sector, it is not surprising that the underlying borrowers have higher concentrations in more expensive housing markets, carry higher loan balances, and are generally more leveraged as compared to their fixed-rate counterparts. Another manifestation of the affordability appeal of hybrid ARMs is that the percentage of purchase loans in alt-A hybrid ARM universe is significantly higher than that in the alt-A 30-year fixed rate sector. Figure 13 highlights the differences between the origination characteristics of alt-A hybrid ARM⁴ and 30-years fixed-rate mortgages from selected shelves.

Figure 13: Comparing Alt-A Hybrid ARMs and 30-year Fixed-rate Collateral

	ARMs	30-yr Fixed
2006 Original Loan Size (\$000s)	356	254
2006 %Conforming	44%	66%
2006 Purchase (%)	61%	47%
2006 Loans with Seconds %	54%	39%
Full Documentation %	27%	35%
California %	43%	32%
Mean Original LTV (%)	73	70
LTV<70%	21%	28%
LTV>=70% and LTV<80%	25%	26%
LTV>=80% and LTV<90%	51%	36%
LTV>=90%	4%	11%

Source: Loan Performance, Banc of America Securities

⁴ Includes 3/1, 5/1, 7/1, and 10/1 hybrid ARMs.

Summary of Collateral Characteristics

Figure 14 collects the statistics for the alt-A 30-year fixed-rate sector presented in our discussion above, while Figures 15 and 16 present these statistics for the alt-A 15-year and alt-A Hybrid ARM sectors, respectively.

Figure 14: Collateral Characteristics for 30-year Alt-A Deals by Issuer Shelf

		Bank Of America BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI	Washington Mutual WMALT
Original Balance (\$bb)	228.8	14.3	104.4	8.6	38.9	53.8	9.0
Original FICO	713	733	712	690	707	717	719
Original LTV	70	67	69	74	68	73	69
Occupancy (%)							
Owner Occupied	80%	50%	86%	83%	86%	73%	82%
Investor	16%	47%	10%	14%	12%	24%	15%
2006 Loan Purpose (%)							
Purchase	47%	51%	48%	54%	39%	49%	46%
Refi. Cashout	34%	30%	34%	34%	36%	32%	39%
Refi Rate/Term	19%	19%	17%	12%	25%	18%	15%
Documentation(%)							
Full Documentation	35%	33%	37%	15%	32%	41%	23%
Property (%)							
Single Family	66%	61%	66%	70%	72%	64%	65%
Geographic Concentrations (%)							
CA	32%	32%	29%	40%	45%	26%	32%
NY	5%	2%	5%	6%	10%	4%	8%
VA	3%	3%	3%	3%	2%	3%	2%
FL	8%	14%	7%	17%	5%	9%	7%
TX	4%	6%	4%	4%	2%	6%	3%
NJ	4%	1%	4%	4%	4%	4%	3%
Rest	44%	42%	49%	27%	32%	48%	44%
2006 Original Loan Size (\$000s)	254	190	258	259	362	227	250
2006 %Conforming	66%	77%	72%	67%	34%	69%	60%
Mean Original FICO	713	733	712	690	707	717	719
FICO<650	10%	3%	11%	24%	12%	6%	6%
FICO >= 650 And FICO< 675	12%	5%	13%	17%	15%	10%	12%
FICO >= 675 And FICO < 700	17%	15%	16%	17%	16%	21%	18%
FICO >= 700 And FICO < 725	18%	19%	17%	14%	15%	20%	19%
FICO >= 725 And FICO < 750	15%	19%	14%	11%	13%	18%	16%
FICO >= 750	25%	38%	25%	15%	21%	25%	28%
FICO N/A	3%	0%	3%	0%	8%	1%	0%
Mean Original LTV (%)	70	67	69	74	68	73	69
LTV<70%	28%	32%	29%	22%	34%	19%	30%
LTV>=70% and LTV<80%	26%	24%	25%	21%	30%	24%	28%
LTV>=80% and LTV<90%	36%	34%	36%	33%	29%	42%	38%
LTV>=90%	11%	9%	10%	25%	7%	14%	3%
Prepay Penalty (%)							
Has Prepay Penalty	9%	2%	4%	61%	19%	4%	14%
Prepay Penalty Unknown	22%	0%	34%	0%	38%	0%	0%

Source: Loan Performance, Banc of America Securities

Figure 15: Collateral Characteristics for 15-year Alt-A Deals by Issuer Shelf

	Aggregate	Bank Of America BOAA	Countrywide CWALT	IMPAC IMM/IMSA	Indy Mac RAST	RFC RALI
Original Balance (\$bb)	18.0	3.1	4.8	0.4	3.3	6.3
Original FICO	721	736	711	692	721	724
Original LTV	58	52	58	63	57	63
Occupancy (%)						
Owner Occupied	71%	28%	80%	80%	87%	76%
Investor	25%	69%	15%	17%	10%	20%
2006 Loan Purpose (%)						
Purchase	20%	20%	20%	27%	19%	21%
Refi. Cashout	41%	38%	41%	48%	33%	45%
Refi Rate/Term	39%	42%	39%	24%	49%	34%
Documentation(%)						
Full Documentation	35%	24%	33%	20%	47%	36%
Property (%)						
Single Family	70%	63%	72%	69%	73%	69%
Geographic Concentrations (%)						
CA	33%	39%	30%	46%	50%	23%
NY	6%	2%	7%	6%	13%	5%
VA	3%	4%	2%	1%	2%	3%
FL	8%	13%	6%	16%	4%	8%
TX	7%	6%	6%	4%	3%	12%
NJ	4%	1%	5%	3%	4%	4%
Rest	39%	36%	44%	24%	25%	46%
2006 Original Loan Size (\$000s)	172	132	162	210	236	188
2006 %Conforming	75%	80%	79%	61%	90%	69%
Mean Original FICO	721	736	711	692	721	724
FICO<650	7%	3%	12%	23%	8%	3%
FICO >= 650 And FICO< 675	9%	6%	14%	17%	10%	7%
FICO >= 675 And FICO < 700	16%	13%	16%	19%	13%	20%
FICO >= 700 And FICO < 725	18%	17%	17%	13%	15%	21%
FICO >= 725 And FICO < 750	16%	18%	13%	11%	16%	19%
FICO >= 750	31%	42%	26%	17%	31%	29%
FICO N/A	2%	1%	2%	0%	8%	1%
Mean Original LTV (%)	58	52	58	63	57	63
LTV<70%	53%	62%	58%	41%	61%	41%
LTV>=70% and LTV<80%	25%	22%	25%	18%	25%	28%
LTV>=80% and LTV<90%	18%	13%	14%	22%	13%	25%
LTV>=90%	4%	3%	3%	15%	2%	6%
Prepay Penalty (%)						
Has Prepay Penalty	5%	4%	1%	58%	12%	2%
Prepay Penalty Unknown	27%	0%	74%	0%	39%	0%

Source: Loan Performance, Banc of America Securities

Figure 16: Collateral Characteristics for Hybrid ARM Alt-A Deals by Issuer Shelf

	Aggregate	Bank Of America BAFC	Bear Stearns BALTA	Countrywide CWALT	Indy Mac INDX	RFC RALI	Lehman SARM
Original Balance (\$bb)	156.7	9.8	47.4	15.8	27.6	14.0	42.1
Original FICO	713	718	707	708	709	716	723
Original LTV	73	71	75	74	72	75	70
Occupancy (%)							
Owner Occupied	81%	83%	76%	84%	87%	82%	83%
Investor	13%	10%	18%	9%	9%	15%	12%
2006 Loan Purpose (%)							
Purchase	61%	59%	67%	65%	55%	60%	57%
Refi. Cashout	26%	23%	22%	21%	31%	27%	29%
Refi Rate/Term	13%	18%	11%	14%	14%	13%	14%
Documentation(%)							
Full Documentation	27%	33%	41%	15%	34%	38%	51%
Property (%)							
Single Family	58%	63%	55%	54%	59%	59%	61%
Geographic Concentrations (%)							
CA	43%	52%	34%	43%	49%	35%	50%
NY	3%	2%	2%	2%	4%	2%	3%
VA	5%	4%	5%	3%	5%	6%	4%
FL	8%	7%	10%	9%	7%	9%	6%
TX	2%	1%	2%	1%	1%	1%	2%
NJ	2%	2%	2%	2%	2%	3%	2%
Rest	38%	32%	45%	40%	32%	43%	33%
2006 Original Loan Size (\$000s)	356	425	318	332	354	317	529
2006 %Conforming	44%	33%	50%	52%	50%	52%	12%
Mean Original FICO	713	718	707	708	709	716	723
FICO<650	8%	6%	10%	11%	10%	5%	4%
FICO >= 650 And FICO< 675	14%	12%	16%	14%	15%	12%	11%
FICO >= 675 And FICO < 700	20%	22%	21%	17%	20%	21%	18%
FICO >= 700 And FICO < 725	19%	17%	18%	19%	18%	20%	20%
FICO >= 725 And FICO < 750	16%	16%	15%	15%	15%	18%	18%
FICO >= 750	24%	28%	20%	21%	22%	24%	30%
FICO N/A	1%	0%	1%	3%	0%	0%	0%
Mean Original LTV (%)	73	71	75	74	72	75	70
LTV<70%	21%	24%	15%	15%	23%	13%	29%
LTV>=70% and LTV<80%	25%	28%	27%	25%	22%	20%	26%
LTV>=80% and LTV<90%	51%	46%	53%	54%	53%	63%	43%
LTV>=90%	4%	3%	4%	5%	3%	4%	2%
Prepay Penalty (%)							
Has Prepay Penalty	28%	12%	28%	26%	34%	18%	31%
Prepay Penalty Unknown	5%	5%	0%	27%	9%	0%	0%

Source: Loan Performance, Banc of America Securities

IV. Prepayment Characteristics of Alt-A Pools

Our issuer-level analysis of collateral characteristics across the alt-A sector suggests substantial variation in origination practices by different issuers, which in turn would suggest distinctions in the prepayment behavior of their shelves. However, taking a step back, our analysis also suggests that the “outside-the-box” loan characteristics, like relatively low loan sizes and lower FICO scores of alt-A borrowers, should lead to less efficient exercise of their refinance option. In fact, a thorough examination of prepayment speeds on alt-A loans leads to two general conclusions:

- Alt-A pools have higher baseline speeds relative to agency and jumbo pools in rate environments where there is no significant incentive to refinance. The primary determinants of the higher baseline speeds are **credit-driven refinancings** (also known as “credit curing”). The fundamental dynamic at work here is the fact that alt-A borrowers can often qualify for a lower rate by improving their credit profile over time. The credit profile of a typical borrower shows gradual improvement as they build up a history of having maintained and managed their debt service, including mortgage payments. A less important determinant of higher baseline speeds on alt-A loans are their higher default rates.
- In the presence of high refinancing incentives, alt-A borrowers prepay at lower speeds relative to Jumbo *and* agency collateral. This diminished responsiveness to refinance opportunities can be attributed to the higher transaction costs, lower credit scores, and the higher loan-to-value ratios associated with alt-A borrowers.⁵

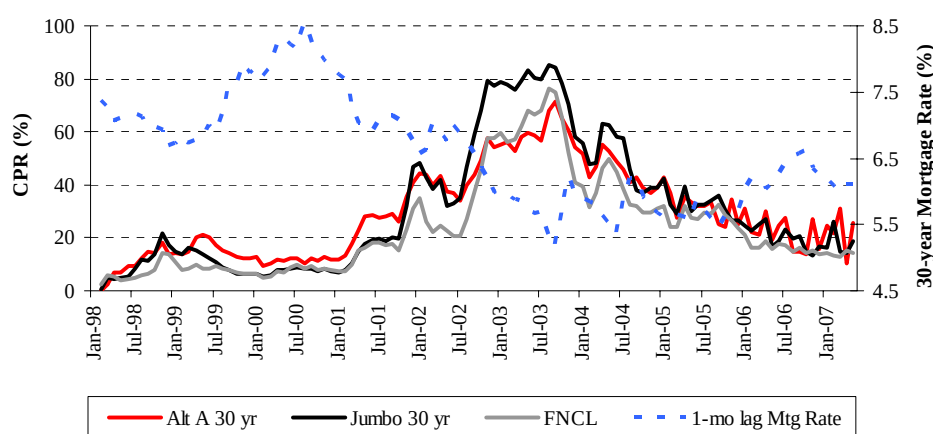
We defer a discussion of how the loan-level characteristics of alt-A borrowers influence these prepayment characteristics to a later section where we analyze issuer specific differences in alt-A prepayment speeds. Our initial focus is to provide a quantitative feel for the differences between alt-A, agency and Jumbo prepayment characteristics. To that end, Figure 17 plots (aggregated) prepayment rates on all 30-year alt-A, agency, and jumbo pools originated in 1998. The motivation behind the choice of the 1998 vintage is that the prepayment rates of this cohort give us a rare opportunity to see what prepayment speeds have been like in out-of-the-money situations on alt-A collateral versus the other sectors. So, for instance, the graph shows that as conventional mortgage rates reached the 8% level in mid-2000, prepayment speeds in all three sectors slowed down to between 8% and 12% CPR. Note that jumbo and agency prepayment rates are virtually indistinguishable over this period, while alt-A prepayment rates are clearly higher (by about 4% CPR in this case). Turning to in-the-money situations, note the differing responsiveness of the three sectors in the graph. In particular, it is important to focus on the changes in prepayment rates across the three sectors, which give you a true idea of the negative convexity of the MBS backed by pools from the different sectors, as opposed to the absolute prepayment levels. Finally, the relatively high speeds on jumbo and alt-A pools in late 2003 can be explained by the presence of 10% clean-up calls in

⁵ For example, a loan-level analysis conducted by S&P showed that an average of 2 points of transaction costs are collected on alt-A loans versus 1 point on jumbo loans. In particular, mortgage holders refinancing investment properties face higher transaction costs.

most non-agency MBS.

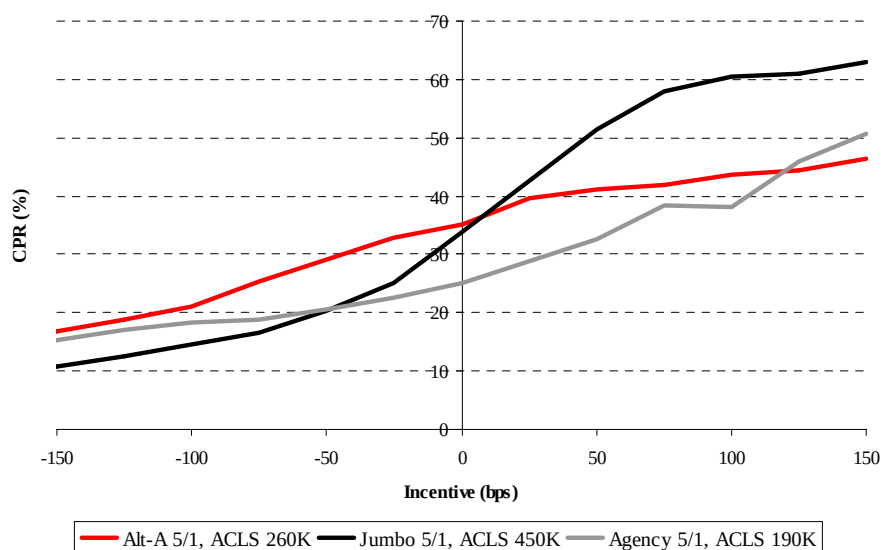
Similarly, alt-A Hybrid ARM collateral also exhibits better convexity than prime jumbo and agency hybrid ARM collateral. Due to the relatively short history of alt-A hybrid ARMs, instead of focusing on any particular vintage, we illustrate this point by simply comparing prepayment S-curves of 12-24 WALA 5/1 Hybrid ARMs from the prime and alt-A sectors based on prepayment data observed since the beginning of 2004 (Figure 18).

Figure 17: Prepayment Speeds on 1998 Vintage 30-year Pools - Alt-A, Jumbo, and Fannie Mae



Source: Loan Performance, Banc of America Securities

Figure 18: Prepayment S-Curves on 12-24 WALA 5/1 Hybrid ARMs: Jan '04 to Present



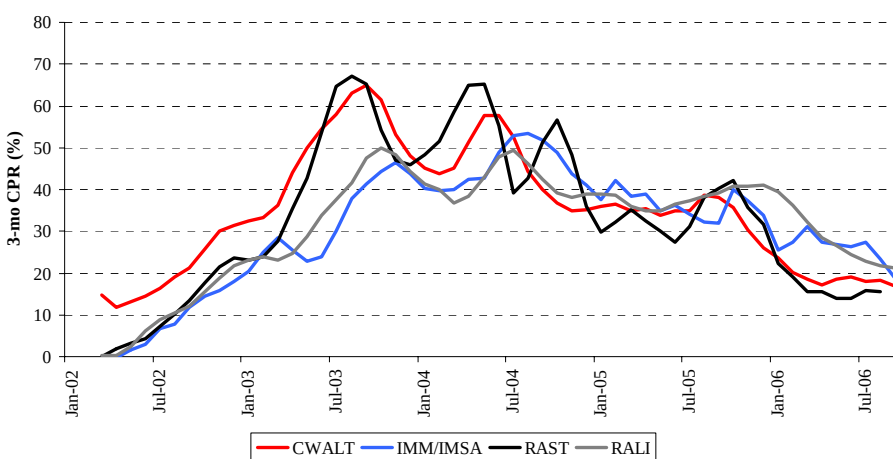
Source: Loan Performance, Banc of America Securities

Issuer-Specific Differences

While the above discussion provides a general feel for how and why alt-A prepayment rates differ from those on agency and jumbo pools, the aggregations used to demonstrate these differences conceal the issuer-specific variability in alt-A speeds. Although the discussion accompanying Figure 3 through Figure 16 alludes to the differences in collateral characteristics across alt-A issuers, it doesn't really provide any insight into how these differences translate into prepayments or, for that matter, as to which of these attributes is most important in influencing prepayment behavior. The following discussion tries to provide some clarity around this important issue.

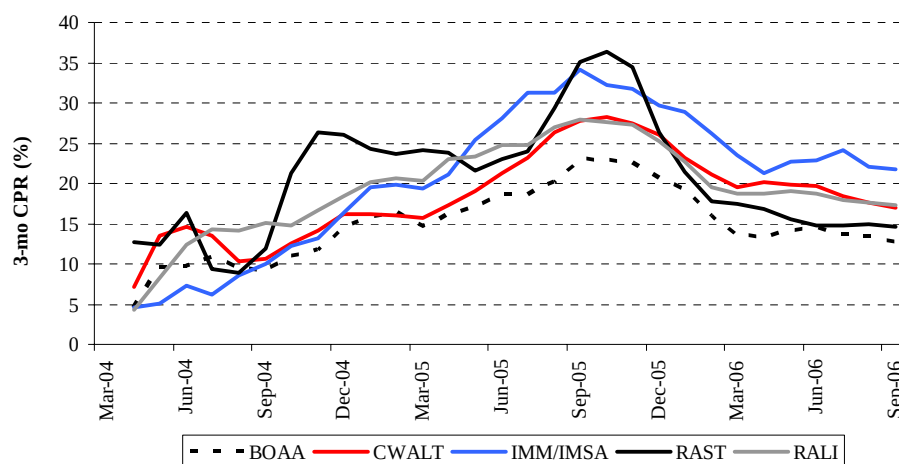
To begin with, it is most useful to review alt-A prepayments disaggregated across issuers, and Figure 19 and Figure 20 provide this data for the 2002 and 2004 originations. The figures highlight the fact that the differences in collateral characteristics among alt-A issuers do, as one might expect, lead to noticeable differences in the prepayment behavior of the different issuers. These differences are perhaps most clearly exposed in Figure 20, which suggests that 5%-10% CPR prepayment differentials during a refinance episode between any two shelves are to be expected and that in certain cases prepayment differentials of 20% to 30% CPR can occur. By contrast, issuer-specific differences tend to be a little more muted in the agency and jumbo sectors.

Figure 19: Prepayment Speeds on 2002 Vintage Alt-A Loans by Issuer



Source: Loan Performance, Banc of America Securities

Figure 20: Prepayment Speeds on 2004 Vintage Alt-A Loans by Issuer



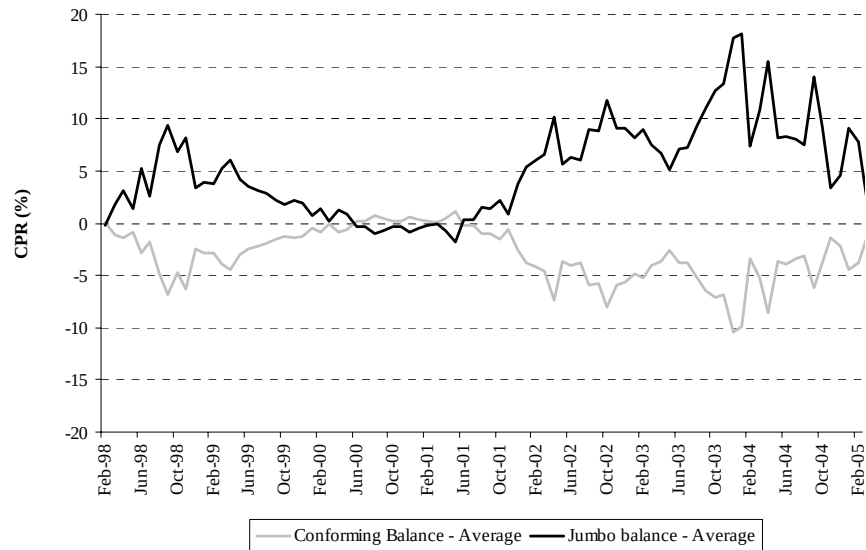
Source: Loan Performance, Banc of America Securities

What are the most important contributing factors to these issuer-level differences? Although a detailed examination of how the loan-level characteristics of alt-A borrowers influence their prepayment characteristics is outside the scope of this paper, a loan-level statistical analysis suggests that the most important factors in determining callability of an alt-A deal are average loan sizes⁶ and the proportion of investor properties. Credit scores, the proportion of high LTV loans, loan purpose, occupancy and documentation are also important, but appear to be second order effects. For example, take a look at Figure 21 and Figure 22. These graphs are constructed by first disaggregating prepayment speeds on the 1998 vintage by different attributes (Conforming Balance versus Jumbo balance, Owner versus Investor) and then subtracting the prepayment rates corresponding to these cuts from the average (the prepayment speed of the entire 1998 vintage). Both graphs show some very interesting trends. In Figure 21, we see that prepayment speeds on Jumbo balance loans are consistently faster (by about 15%-20% CPR) than those on conforming balance loans during refinancing episodes. On the other hand, notice the convergence in prepayment speeds on the two types of loans in 2000 when prepayment speeds were dominated by turnover. In Figure 22, we see that investor properties refinance less efficiently than owner-occupied properties and that the magnitude of the difference can be as much as 15%-20% CPR during refinance waves. Turning to the 2000 experience, we see that baseline speeds on investor properties were as much as 3%-5% CPR higher than on owner-occupied properties. The difference between the prepayment speeds on the two different property types in these “out-of-the-money” scenarios appears to be primarily driven by credit-driven refinancings. In general, investor loans in alt-A pools carry a premium mortgage rate because of their high LTVs or documentation issues. However, the passage of time cures many of these hurdles for the borrower and allows them to qualify for a better mortgage rate. Figure 22 is a microcosm of the alt-A experience – the high number of investor properties in alt-A pools is an important determinant of why

⁶ We construct a “rule of thumb” metric for the loan size effect by computing the percentage of conforming balance loans in a given alt-A pool.

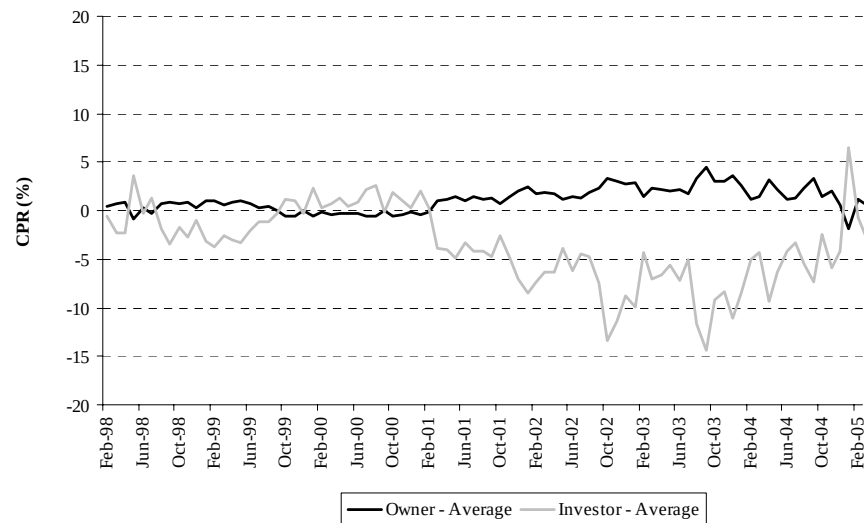
baseline speeds on alt-A pools are higher than those on agency and jumbo pools.

Figure 21: Relative Prepayment Speeds on 1998 Vintage 30-year Pools by Loan Size



Source: Loan Performance, Banc of America Securities

Figure 22: Relative Prepayment Speeds on 1998 Vintage 30-year Pools by Occupancy Type



Source: Loan Performance, Banc of America Securities

These two attributes (proportion of conforming balance loans, proportion of investor properties) can explain most of the differences between alt-A issuers seen in the above figures. Figure 23 provides one piece of evidence for this claim by rank ordering the prepayment speeds shown in Figure 20 by originator. Note that Bank of America's alt-A shelf, BOAA, which has the lowest 6-month prepayment speed, also has both the highest percentage of investor properties (53%) and conforming balance loans (83%). In

fact, if we ignore the results from the IMPAC shelf, which are distorted by the extremely high level of prepayment penalties, we see that the ordering is heavily correlated with the proportion of conforming balance and investor loans.

Figure 23: Selected Collateral Characteristics of 2004 Alt-A Originations⁷ by Issuer (as of Sep 2006)

Shelf	% Conf Balance	% Investor	% Prepay Penalty	6-mo %CPR
BOAA	83	53	10	13.6
RAST	39	9	16	14.7
RALI	68	24	16	18.0
CWALT	79	17	2	18.3
IMM/IMSA	61	20	60	22.3

Source: Loan Performance, Banc of America Securities

Dependence of Prepayment Speeds on HPA

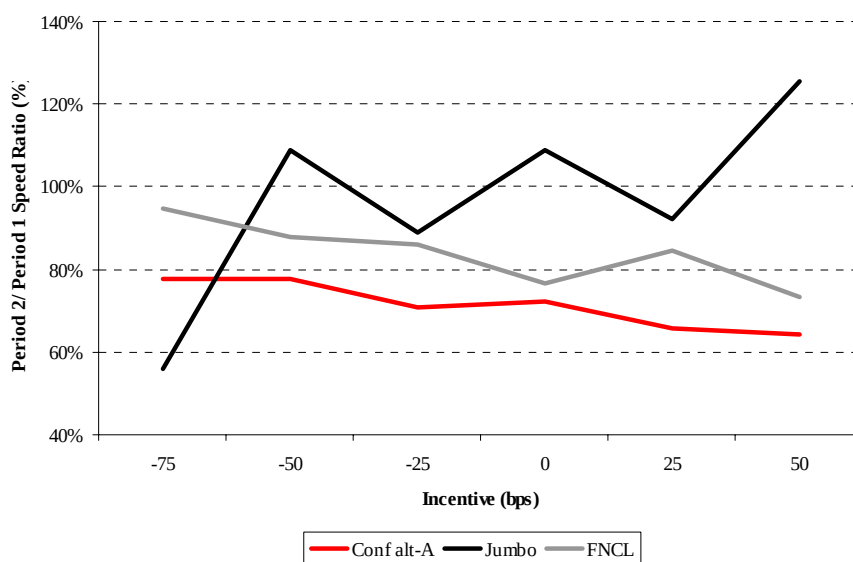
The recent slowdown in the housing market provides a good opportunity to see how a slowdown in HPA affects prepayments across the different mortgage sectors. Figure 24 plots (as a ratio) prepayment S-curves for 30-year jumbos, conforming alt-As and FNCLs observed in two periods – December 2005 to May 2006 and December 2006 to May 2007. The reason for choosing these two periods is because speeds in the three sectors started to diverge around November 2006. The figure shows that speeds on conforming alt-As and FNCLs have slowed down dramatically across the two periods with alt-As slowing down a lot more than FNCLs. Speeds on Jumbos, however, have barely changed.

The reason speeds on Jumbos have not slowed down over the last couple of months is because unlike the alt-A or conventional fixed-rate borrowers, these borrowers have low LTVs and are not strapped for cash. Hence, they do not extract equity as often as alt-A or conventional borrowers, making them less leveraged to HPA. At the same time, since Jumbo borrowers usually have pristine credit characteristics, they already have the best available mortgage rate reducing the possibility of credit curing. Alt-A borrowers on the other hand are assisted by a strong HPA environment since the increased equity makes it easier to cashout or refinance into a new mortgage loan with a lower rate. Both these refinancings become more challenging as the housing market cools off.

To see the relative impact of HPA on Jumbos, alt-As and FNCLs from a different perspective, we plot the difference in prepayment S-curves on fixed-rate loans originated in High and Low HPA states in Figure 25. We have only considered speeds since the beginning of 2006 to plot these S-curves. As expected, the alt-A borrowers are most leveraged to HPA with speeds in the low HPA states being significantly slower than speeds in the high HPA states. FNCLs are less leveraged but still show a noticeable slowdown in speeds in the low HPA states whereas the effect of HPA on Jumbo prepay speeds is minimal.

⁷ Note that a substantial percentage of loans backing the RALI and CWALT deals do not have prepay penalty data.

**Figure 24: The Evolution of the S-Curve on Various 30-Year Mortgage Sectors:
Dec '05 – May '06 versus Dec '06 – May '07***

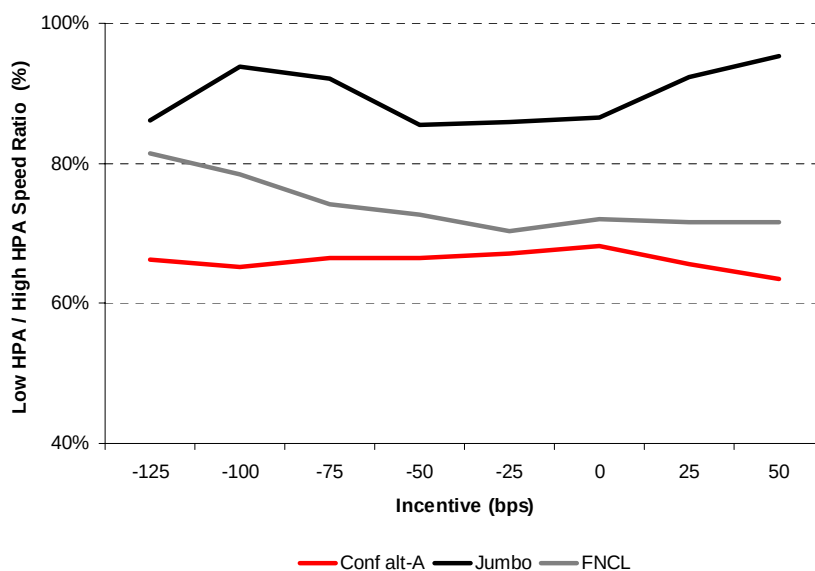


* All pools have a WALA between 12-24 months

** Period 1: Dec 05 – May 06; Period 2: Dec 06 – May 07

Source: Loan Performance, Banc of America Securities

Figure 25: S-curves for 30-Year Jumbos, Alt-As and FNCLs in Low and High HPA States



* All pools have a WALA between 12-24 months

Source: Loan Performance, Banc of America Securities

V. Credit Performance in the Alt-A Sector

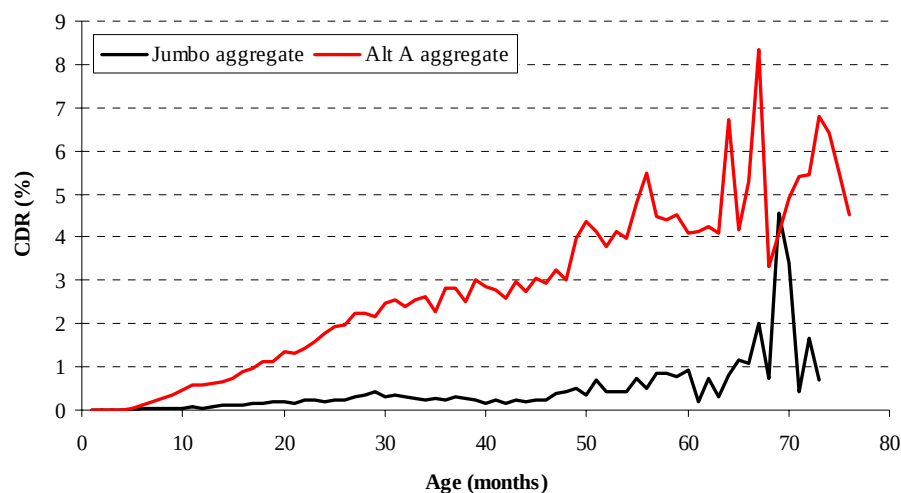
We have divided our discussion on the credit performance in the alt-A sector into three parts. First, we take a look at historical losses in the alt-A fixed and hybrid ARM sectors relative to the losses in the corresponding sectors in the jumbo market. Next, we analyze the impact of different loan characteristics and the strength of the housing market on credit performance of alt-A mortgages. Finally, we analyze some seemingly simple trends in mortgage delinquencies to show the complexity that lies beneath. The recent downturn in mortgage credit performance has led to market observers making sweeping generalizations about the alt-A and subprime sectors but our analysis shows that there is considerable variation in delinquency rates at the issuer and shelf levels.

Historical Credit Losses in alt-A and Jumbo Mortgage Markets

Figure 26 compares default rates on Jumbo and alt-A pools across recent vintages and shows that the default curve for alt-As is both steeper and higher. This doesn't come as a big surprise as two of the key determinants of default rates on mortgages are LTV ratios and credit scores and, as documented in the discussion accompanying Figure 4 and Figure 7, alt-A borrowers have higher OLTVs and lower credit scores.

Of potentially more interest to investors in the lower rated tranches of alt-A deals are Figure 27 through Figure 29, which compare cumulative losses on jumbo and alt-A loans for 30-year and 15-year fixed-rate and hybrid ARM sectors. As was the case with default curves, the cumulative loss curve for alt-A loans appears to be both steeper and higher than on jumbo loans.

Figure 26: Default Rates on 30-year Jumbo and Alt-A Collateral



Source: Loan Performance, Banc of America Securities

Figure 27: Cumulative Losses in Basis Points on 30-year Jumbo and Alt-A Collateral (as of July 2007)

Vintage Year	Jumbo	alt-A
1995	12.1	81.9
1996	6.9	30.9
1997	3.7	25.8
1998	2.8	20.2
1999	4.8	59.0
2000	5.4	69.8
2001	1.7	47.6
2002	2.4	43.8
2003	1.7	18.7
2004	0.5	7.6
2005	0.4	2.3
2006	0.0	0.3

Source: Loan Performance, Banc of America Securities

Figure 28: Cumulative Losses in Basis Points on 15-year Jumbo and Alt-A Collateral (as of July 2007)

Vintage Year	Jumbo	alt-A
1995	4.9	12.3
1996	2.3	8.0
1997	0.5	4.8
1998	0.7	7.7
1999	1.0	43.2
2000	0.9	31.8
2001	0.3	6.2
2002	0.2	3.7
2003	0.1	2.3
2004	0.2	1.8
2005	0.0	0.3

Source: Loan Performance, Banc of America Securities

Figure 29: Cumulative Losses in Basis Points on Hybrid ARM Jumbo and Alt-A Collateral (as of July 2007)

Vintage Year	Jumbo	alt-A
2000	3.2	43.0
2001	1.5	11.6
2002	1.5	9.1
2003	1.9	9.5
2004	1.3	5.7
2005	0.8	3.2
2006	0.0	0.7

Source: Loan Performance, Banc of America Securities

Factors Impacting Credit Losses on Alt-A Collateral

In general, credit losses on mortgage pools depend upon a host of variables including macroeconomic factors (home price appreciation (HPA), interest rates, unemployment, personal income etc.), collateral characteristics (FICO, LTV, Occupancy Type, Loan Purpose, Documentation Type etc.), underwriting standards, and borrower liquidity (cash reserves) among others. The issue is further complicated in the alt-A sector by a significant amount of deal-level variability in terms of collateral/borrower characteristics and underwriting standards. The final nuance to consider is that even given a loss forecast for alt-A loans, the impact of these losses at deal-level can be quite different depending upon how the deal is structured (i.e., shifting-interest versus OC structures) and the timing of the loss.

To facilitate the discussion on the impact of different collateral characteristics on credit losses of mortgage pools, the origination characteristics of different alt-A vintages are shown in Figure 30, while cumulative losses on 30-year fixed-rate alt-As as a percentage of the original balance are shown in Figure 31. As Figure 31 makes clear, the 2000 and 2001 alt-A vintages have performed a lot worse than the 2002-2005 cohorts in terms of cumulative losses. In general, the 2000 cohort was originated in a purchase environment where mortgage originators lowered their lending standards along the following dimensions in order to generate higher origination volumes:

- For the 2000 alt-A vintage, the average FICO score was slightly lower and the percentage of borrowers with FICO score less than 680 was somewhat higher relative to other cohorts.
- The original LTVs for the 2000 vintage were higher than those for the surrounding years.

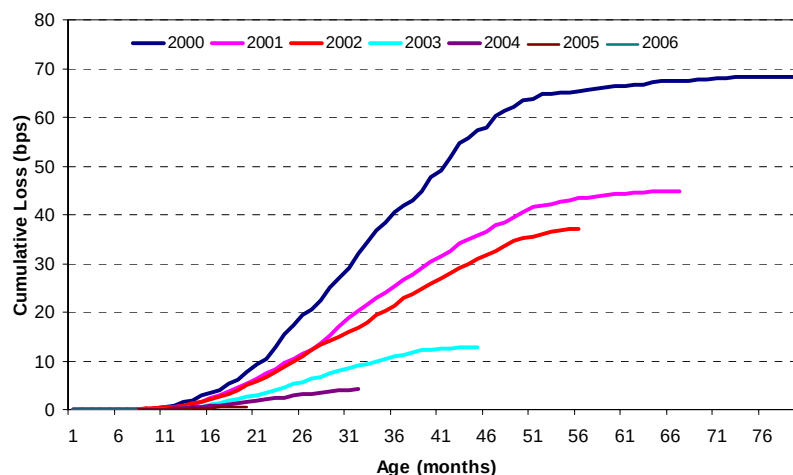
In addition, the percentage of purchase borrowers was higher in 2000 which in and of itself seems to be associated with somewhat higher loss levels as discussed later in this section. Some improvement in collateral characteristics in conjunction with a very high HPA environment and a heavy refinancing wave resulted in lower cumulative losses on 2002-2004 alt-A cohorts. In this section, using 2000-2001 vintages as the test cases, we analyze the impact of different loan characteristics and the strength of the housing market on the credit performance of mortgages.

Figure 30: Origination Characteristics of Different Alt-A Cohorts

	Year	AOLS ('000')	FICO	% FICO < 680	% Cashout	% Purchase	% Full Doc	% OLTV	% Seconds	% IO	Original Balance(MM)
Hybrid ARM	2000	181	683	38	24	71	61	73	0	8	891
	2001	352	707	30	36	50	30	68	2	24	4,451
	2002	329	706	31	33	48	28	70	1	24	12,151
	2003	318	707	29	30	49	31	71	22	53	25,262
	2004	272	709	28	25	61	33	74	38	72	102,457
	2005	282	712	26	26	63	30	73	43	80	125,858
	2006	323	710	28	25	64	22	73	53	85	70,376
Fixed	1998	154	712	23	28	46	40	72	0		19,903
	1999	151	699	34	29	51	40	74	0	0	12,541
	2000	170	697	34	21	67	36	78	0	0	10,385
	2001	256	705	29	28	54	33	76	0	0	19,724
	2002	230	707	30	27	52	37	75	1	0	32,564
	2003	214	711	27	34	41	38	70	7	2	48,531
	2004	196	711	27	32	51	35	71	15	8	51,250
	2005	220	716	24	39	47	39	69	25	30	99,366
	2006	242	710	30	36	51	24	70	34	40	71,784

Source: Banc of America Securities

Figure 31: Cumulative Losses on Fixed-rate Alt-A Loans by Vintage Year

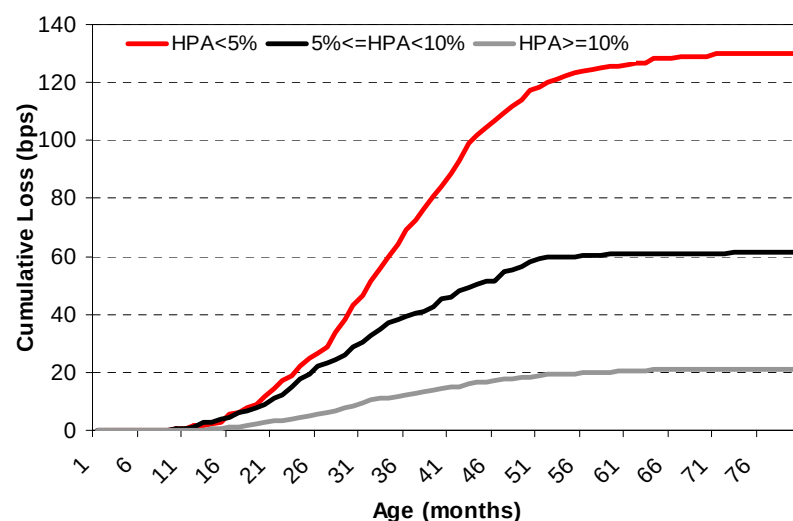


Source: Banc of America Securities

Alt-A Credit Performance in Different HPA Environments

Figure 32 plots cumulative losses on 2000 and 2001 30-year fixed-rate alt-A loans as a function of the (annualized) HPA. The figure shows that losses on alt-A mortgages that experienced less than 5% HPA were more than double the losses on mortgages that experienced 5%-10% HPA, and nearly 6 times the losses on the pools that experienced higher than 10% HPA. The strong dependency of losses on HPA is an issue of some concern for the market at the moment because the 2006 origination cohort is likely to experience substantially lower HPA than the 2000 and 2001 cohorts, which themselves were the worst alt-A cohorts from credit perspective over the past few years.

Figure 32: Cumulative Losses of 2000-2001 Fixed-rate Alt-A Loans by Annualized HPA



Source: Banc of America Securities

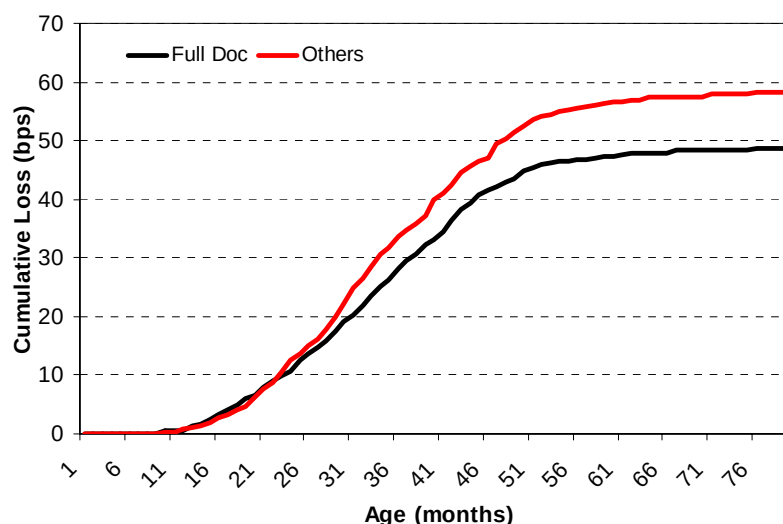
Alt-A Credit Performance Based on Documentation, Loan Purpose and Loan Size

The significance of LTV and credit score from a credit performance standpoint cannot be overstressed. In this section, we focus on the historical dependence of losses on some less obvious factors including documentation level, loan purpose and loan size that also played a key role in determining the credit performance of 2000-01 vintages. These three factors are also very relevant in the current environment since the recent cohorts have very low percentage of borrowers with “full documentation,” a higher fraction of purchase loans, and considerably higher average loan sizes.

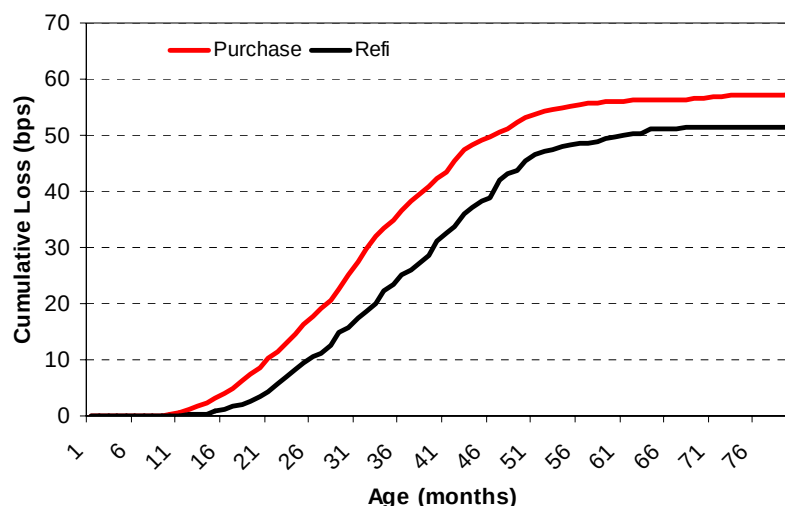
Figure 33 shows cumulative losses of 2000 and 2001 cohorts for full-documentation and low-documentation loans. The cumulative losses on full documentation loans were about 10 bps lower than the losses on low and no-documentation loans. Similarly, Figure 34 shows losses on purchase and refi mortgages from 2000-2001 cohorts. In general, purchase loans have slightly higher losses than refi loans.

Finally, loan size can play an important role in determining cumulative losses as the fixed-costs associated the foreclosure process lead to higher losses as a fraction of the original loan size when the original loan size is low. As shown in Figure 35, mortgages with loan sizes greater than \$200K had about 40 bps lower losses than losses on lower loan balance mortgages for the 2000-2001 vintage. However, it is worth pointing out here that the higher loan balance mortgages from the 2000-2001 vintage were disproportionately concentrated in California, which experienced a booming housing market in the following years. The concentration of higher loan balance loans in California accounts for a portion of the drop in losses experienced by higher loan size bucket.

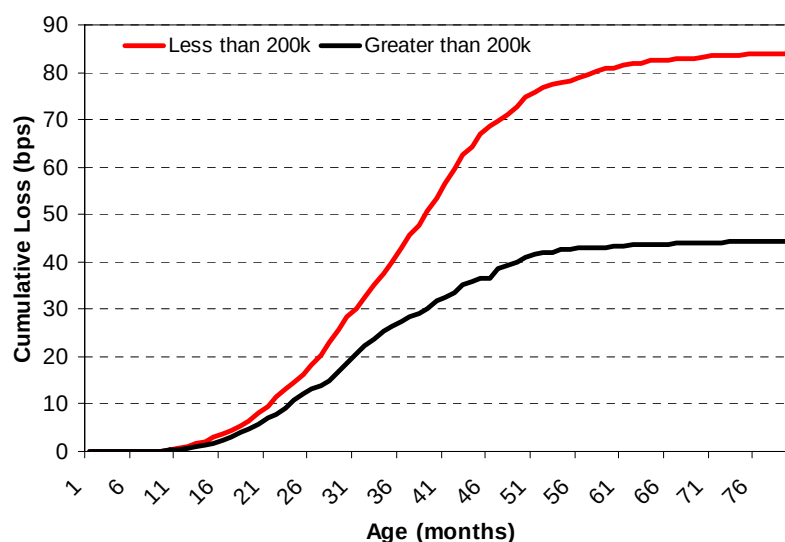
Figure 33: Cumulative Losses of 2000-2001 Fixed-rate Alt-A Loans by Documentation Level



Source: Banc of America Securities

Figure 34: Cumulative Losses of 2000-2001 Fixed-rate Alt-A Loans by Loan Purpose

Source: Banc of America Securities

Figure 35: Cumulative Losses on 2000-2001 Fixed-rate Alt-A Loans by Loan Size

Source: Banc of America Securities

Deal Level Variation in Credit Losses

We show the average cumulative losses and standard deviation of losses at deal level for 2000 and 2001 fixed-rate vintages in different HPA ranges in Figure 36. (Note that the cumulative losses shown here are slightly different from the numbers shown in Figure 31 because the deal level analysis is done using issuance dates while cohort losses are calculated using the origination date.) It is interesting to note that not only the average level of loss is strongly dependent on HPA, but the deal level variability in losses also increases with a decrease in HPA. This result indicates that in weaker housing markets underwriting standards and/or origination characteristics become a lot more important in determining credit performance.

Figure 36: Mean and Standard Deviation of Losses in Basis Points on 2000-2001 Vintage Alt-A Fixed-rate Mortgages

Year		HPA Bucket			Total
		<5%	5-10%	>10%	
2000	Mean	253.2	84.9	22.7	74.8
	St. Dev.	269.7	96.1	27.1	79.1
2001	Mean	108.6	56.2	13.2	48.2
	St. Dev.	114.4	67.0	18.0	44.6

Source: Banc of America Securities

The Simplicity of Residential Mortgage Credit

The recent downturn in mortgage credit performance has led to market observers making sweeping generalizations about the alt-A and subprime sectors but the discussion presented in the previous section indicates that there are several variables that impact mortgage credit performance. In fact, a close look at alt-A delinquency rates at the issuer and shelf levels shows that there is a significant variation in the credit performance of loans categorized as “alt-A” depending upon who originated these loans. In this section, we analyze the volatility in delinquency rates on loans across different sectors as well as at the issuer and deal levels for alt-A mortgages.

Residential Non-Agency Mortgage Credit Performance by Sub-sector

In a nutshell, among other factors, mortgage credit performance depends on a number of different collateral characteristics including credit scores, borrower LTVs, documentation type, origination channel etc. At the end of the day, the originator distills all these factors into the mortgage rate that they charge the borrower. Borrowers who are perceived to pose a greater credit risk are charged higher mortgage rates. Therefore, one simple way to capture the credit risk posed by any individual loan is to capture the difference (“spread”) between this rate and the rate available to borrowers with the best credit at the moment the loan was originated. This metric is known as spread-at-origination (SATO). Our approach to solving the “curse of multi-dimensionality” and depicting the diversity of mortgage credit performance is thus to show the delinquency rate of loans from the different non-agency sectors versus their SATOs.

The “bubble” graphs shown in Figures 37 and 38 plot the average 60+ delinquency rate for fixed- and adjustable-rate loans from different non-agency sectors versus their SATOs. Some features of the graph deserve explanation. First, we have only selected performance data corresponding to loans with 6-8 months of seasoning from the 2006 vintage. This controls for the dependence of delinquency rates on seasoning. Second, loans from the different sectors (jumbo, alt-A, subprime) are colored different (red, blue, purple). Finally, each of the bubbles in the figure represents loans (aged 6-8 months) from a particular non-agency shelf. A larger bubble represents a non-agency securitization shelf with a larger population of such loans.

Figures 37 and 38 show a number of interesting trends:

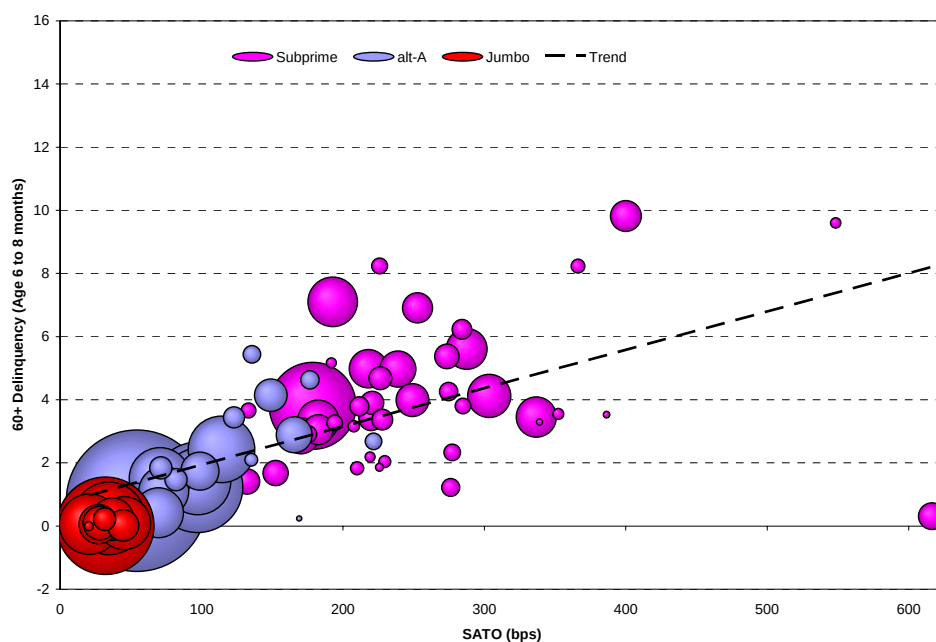
- There is a significant amount of variation in credit performance across the different sectors with 60+ delinquency rates ranging from greater than 10% to less than 0.5%.
- The volatility of the credit performance of different shelves in the individual

sub-sectors increases as we go down in credit. Credit performance on the prime jumbo shelves is tightly clustered together because of the relative uniformity of the underwriting practices employed by the different originators. Variation in performance increases somewhat as we move down-in-credit to the alt-A sector, and significantly as we move to the subprime sector.

- The recent issue of how much overlap there is between the subprime and alt-A sectors is nicely resolved by the figure. Clearly, some of the lower-tier shelves in the alt-A sector appear to be performing on par with some of the “best” subprime shelves but the relative size of this loan population is a relatively small part of the alt-A sector.
- The risk-based underwriting principles used by originators appear to be relatively rational with loans higher SATOs showing worse credit performance.

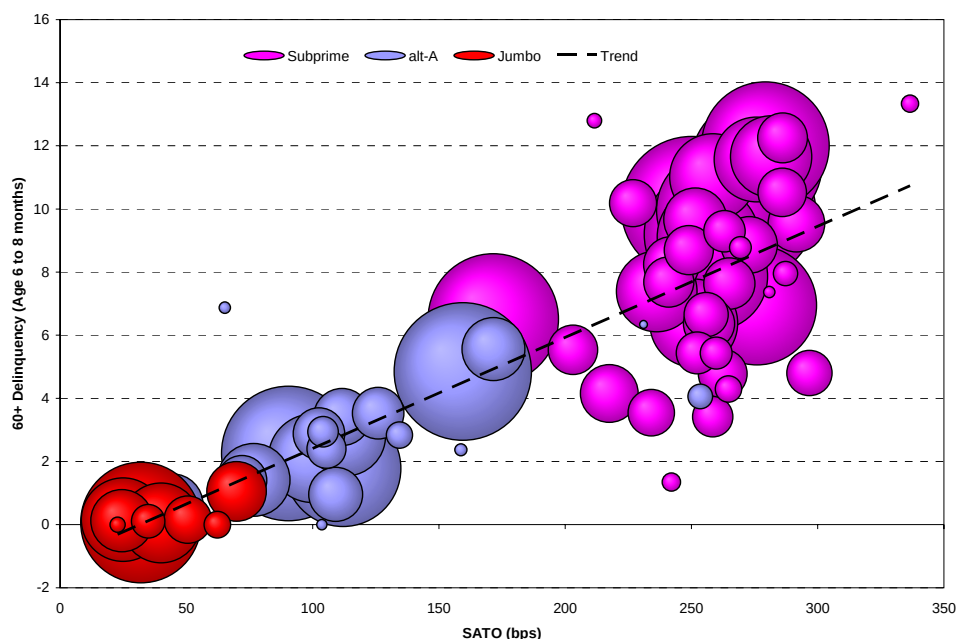
A couple of caveats are worth noting. We are not claiming that SATO represents a perfect measure for capturing all the nuances of mortgage credit and our analysis sidesteps the issue of whether the credit risk in higher SATO loans is appropriately priced in the market.

Figure 37: Credit Performance Across Various RMBS Sub-sectors for Fixed-rate Mortgages*



*As of January 2007. 60+ Delinquencies are expressed as a percentage of the outstanding balance.

Source: LoanPerformance, Banc of America Securities

Figure 38: Credit Performance Across Various RMBS Sub-sectors for Adjustable-rate Mortgages*

*As of January 2007. 60+ Delinquencies are expressed as a percentage of the outstanding balance.

Source: LoanPerformance, Banc of America Securities

The Simplicity of Alt-A Credit Performance

In keeping with our theme of simplicity, we now magnify the segment of Figures 37 and 38 that represent credit performance on alt-A loans. The results, at varying levels of resolution, are collected in Figures 39 through 42.⁸ Focusing on the tables presented in Figures 39 and 40 first, the number of columns in the tables eloquently speaks to the “multi-dimensionality” issue described above: each of collateral characteristics listed can be shown to be correlated with credit performance. Still, the tables provide a useful perspective on how diverse the underwriting standards in the alt-A sector are. For example, let’s focus on the shelf-level variation in the number of adjustable-rate loans (see Figure 40) that have seconds, to highlight one of the characteristics that has received a fair amount of attention recently. The numbers range from 0% for the FHAMS shelf to 70% for the DBALT shelf. Perhaps unsurprisingly, the 60+ delinquency rate on loans from these two separate shelves average 0.96% and 3.53% respectively. Turning to the fixed-rate loans (see Figure 39), the BOAA shelf has only 2% IO loans and a delinquency rate of 0.51% compared to the IMSA shelf with 66% IO loans and a delinquency rate of 5.44%.

Figure 41 simplifies the complexity of the multitude of different collateral characteristics presented in Figures 39 and 40 by graphing 60+ Delinquency rates by SATO as before. Once again, even at its somewhat lower level of resolution, the bubble graph provides an easy way to visualize the dispersion in delinquency rates across different shelves and the relative importance of each of these shelves. As even a casual inspection of the figure shows, the larger alt-A issuers are still showing relatively good

⁸ Our naming convention for the shelves is based on Bloomberg naming conventions.

credit performance compared to the smaller originators for the 2006 vintage (on an absolute basis the 2006 vintage is much worse than earlier vintages). The trouble appears to lie more with the relatively smaller population of alt-A loans with high SATOs.

Figure 42 focuses on loans currently aged 6-8 months from different deals in a Countrywide shelf. Familiar themes appear once again: there is significant variation in credit performance and credit performance is correlated to SATO.

Figure 39. Collateral Characteristics of the 2006 Alt-A Fixed-rate Vintage by Shelf

									%	%						60+ Del.
	Original	AOLS				%	%	%	FICO	FICO	%	%	%	%	%	(Age 6-8
Shelf	(\$MM)	WAC	(000)	FICO	CLTV	Cashout	Purchase	Investor	(<680)	(<620)	Seconds	IO	Full Doc	HasPenalty	CA	mos)
ARMT	255	7.85	209	695	75	26	65	34	39	0	0	30	7	24	7	4.62
AHM	122	7.51	202	694	83	34	57	21	39	3	63	59	15	17	10	2.10
BOAA	2,880	6.75	185	724	69	32	55	29	17	0	0	2	32	0	20	0.51
BSABS	345	7.35	216	693	84	30	62	24	41	0	65	36	18	22	12	3.44
CSMC	1,853	6.84	276	710	73	33	51	16	28	0	45	35	16	24	23	1.10
CMLTI	1,026	7.78	179	675	79	50	42	22	63	0	7	12	0	56	16	2.89
CWALT	23,528	6.72	250	713	71	36	49	7	27	0	36	39	26	8	26	0.81
DBALT	827	7.63	214	697	84	32	60	30	37	0	60	61	10	43	21	4.14
FHAMS	2,539	6.91	221	722	65	39	50	23	21	0	0	40	27	2	17	0.43
GSAA	1,088	7.05	226	702	79	38	51	20	36	0	52	55	16	33	24	1.74
IMSA	244	7.50	289	691	78	40	52	23	45	3	49	66	6	78	44	5.44
RAST	3,161	6.86	276	702	75	46	40	10	36	0	42	39	21	62	35	1.51
LXS/LMT	4,664	7.34	245	696	78	36	51	19	41	0	34	44	21	36	22	2.43
MSM	360	6.79	259	706	73	43	40	18	32	1	46	47	14	57	35	1.84
RAAC	23	7.71	165	686	82	42	47	29	52	14	20	16	61	44	9	0.24
RALI	12,819	7.13	228	710	78	32	52	19	27	0	44	42	25	5	18	1.22
RAMP	337	8.40	133	691	100	32	59	24	45	13	0	14	77	72	1	2.68
TBW	387	6.94	201	704	79	39	37	22	31	0	42	44	15	6	9	1.47
WMALT	3,917	7.09	264	710	78	37	48	16	28	0	49	48	15	18	27	1.42
Aggregate	60,379	6.96	238	710	74	36	50	15	29	0	36	39	24	16	23	1.34

*As of January 2007

Source: Loan Performance, Banc of America Securities

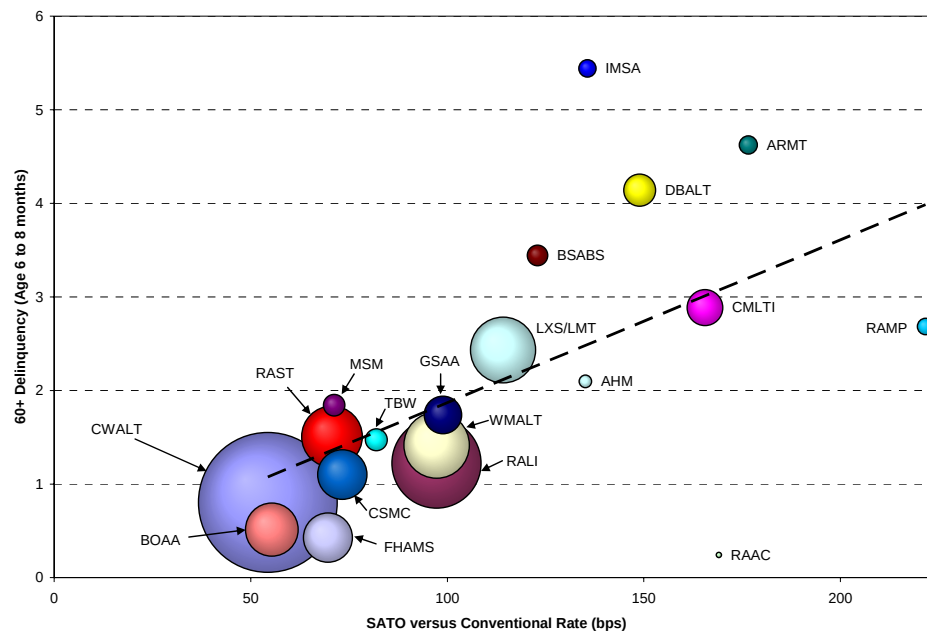
Figure 40. Collateral Characteristics of the 2006 Alt-A Adjustable-rate Vintage by Shelf

	Original	AOLS				%	%	%	%	%	%	%	%	%	%	60+ Del.
Shelf	(\$MM)	WAC	(000)	FICO	CLTV	Cashout	Purchase	Investor	FICO	FICO	%	%	Full	%	%	(Age 6-8
									(<680)	(<620)	Seconds	IO	Doc	HasPenalty	CA	mos)
ARMT	1,471	6.87	388	718	74	24	65	15	22	0	15	83	18	35	38	3.40
AHM	68	7.10	268	711	80	17	68	30	22	0	71	84	16	14	17	2.37
BAFC	1,054	6.40	418	711	79	28	58	7	24	0	49	93	27	9	50	1.40
BALTA	8,370	7.27	289	704	87	20	74	21	31	0	73	90	12	43	31	4.85
CMLTI	1,621	6.32	401	729	81	25	65	7	13	0	54	92	4	4	45	0.62
CWALT	5,032	6.86	307	706	82	25	64	12	31	0	58	89	21	41	43	2.09
DBALT	1,177	6.97	309	710	86	21	68	13	24	0	70	91	13	59	49	3.53
FHAMS	2,025	6.94	267	721	70	29	61	27	18	0	0	86	30	0	18	0.96
GSAA	6,371	6.83	284	710	84	22	68	14	27	0	62	77	40	31	38	1.77
GSCC	769	6.63	355	723	83	20	71	8	16	0	67	82	15	22	48	2.40
HVMLT	404	6.78	630	704	81	27	59	11	31	0	51	93	11	36	63	2.94
IMSA	298	7.01	423	712	77	29	66	55	26	1	46	59	34	91	46	2.84
INDX	9,348	6.70	334	707	79	30	59	9	30	0	52	89	17	51	50	2.26
LXS	3,407	7.50	372	702	88	21	71	16	33	0	62	90	16	44	32	5.55
MSM	1,147	6.71	426	710	78	34	53	15	30	0	33	91	20	33	42	2.88
RAAC	27	7.92	201	674	91	21	69	15	51	21	44	45	44	48	9	6.34
RALI	4,980	6.94	320	711	84	23	62	14	25	0	61	87	21	21	35	2.80
RAMP	373	8.38	154	678	100	23	73	21	50	20	0	19	74	72	3	4.06
RFMSI	283	6.48	471	742	75	28	49	0	6	0	43	87	57	0	42	0.00
SAMI	68	5.73	420	724	82	15	77	7	15	0	55	0	12	23	54	6.87
SARM	3,820	6.62	512	726	77	28	56	8	13	0	48	92	19	40	54	1.41
WMALT	976	6.71	317	719	72	32	48	22	16	0	25	87	20	31	50	0.00
Aggregate	53,090	6.90	327	710	82	25	64	14	27	0	55	87	21	37	40	2.80

*As of January 2007. Option ARMs are not included.

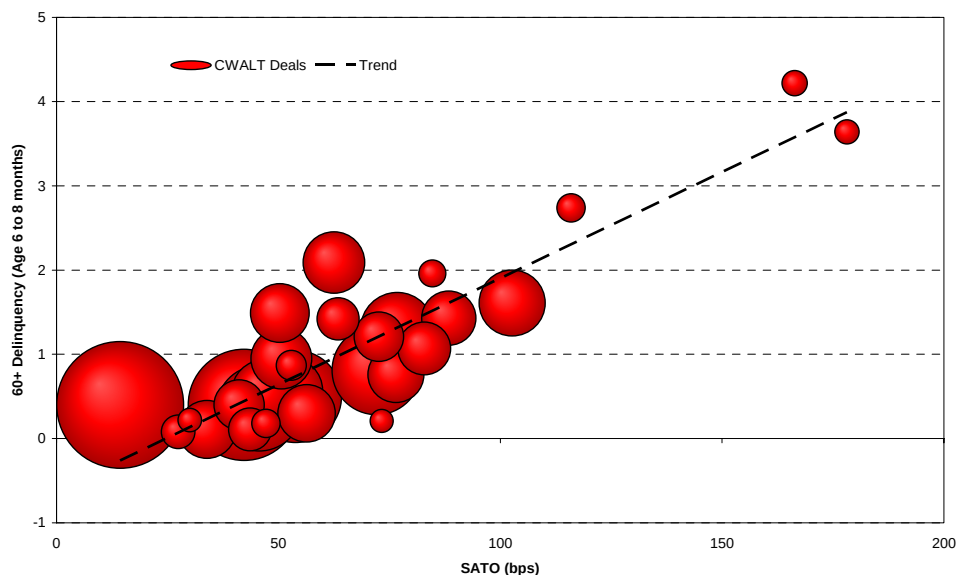
Source: Loan Performance, Banc of America Securities

Figure 41: The Simplicity of Fixed-Rate Alt-A Credit Performance



*as of January 2007. All data for the 2006 Vintage. 60+ Delinquencies are expressed as a percentage of the outstanding balance.
Source: Loan Performance, Banc of America Securities

Figure 42: Credit Performance for Loans Aged 6-8 Months for Different Deals on the CWALT Shelf



*as of January 2007. All data for fixed-rate loans from the 2006 Vintage. 60+ Delinquencies are expressed as a percentage of the outstanding balance.
Source: Loan Performance, Banc of America Securities

A Simple Takeaway

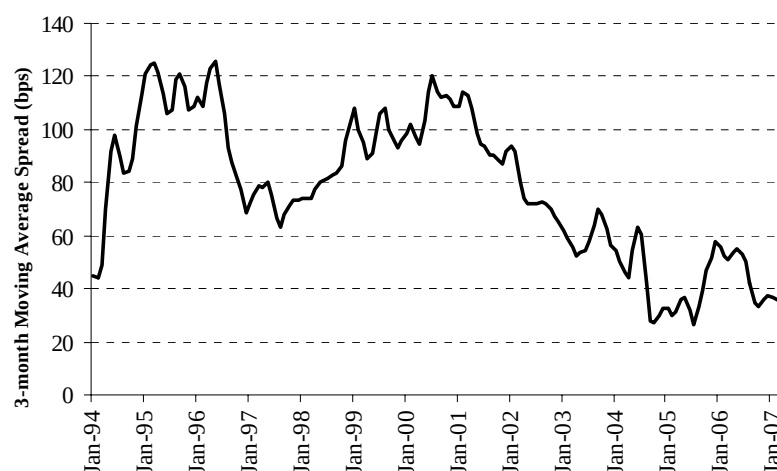
At higher levels of magnification, the variations in mortgage credit performance at the sector, shelf and deal level suggest the importance of avoiding sweeping generalizations. As is often true, differentiation (prudent security selection) makes a difference.

VI. The Evolution of the Alt-A Sector

As with any new market, the alt-A sector has evolved since its inception as the mix of participating lenders has changed and investors have become more comfortable with the alt-A “story.” However, the future of the sector has been a source of debate for some years because of the increasing involvement of the GSEs in the alt-A sector and, more recently, because of a severe downturn in mortgage credit which has resulted in increased delinquency rates on alt-A pools. In terms of the first issue, over the years the GSEs have continued to reshape their underwriting guidelines to accommodate a broader mix of borrower credit profiles. This involvement has a number of negative implications for the private-label sector. First, with their access to cheap capital, the GSEs can compete effectively with the private-label sector for alt-A loans and can thus affect liquidity in the sector. Next, their competitive presence would potentially compress the spread over conforming rates paid by the alt-A borrowers making existing alt-A pools more refinaneable. Finally, the GSEs would siphon away conforming balance borrowers leading to alt-A pools with a more negatively convex profile.

To a certain extent, all the three encroachments listed above occurred in the alt-A marketplace. For example, as Figure 43 shows, the spread between alt-A and conforming mortgage rates narrowed meaningfully between 2000 to 2005. Over this period, a significant part of the spread compression appears to have resulted from alt-A lenders targeting borrowers with better credit profiles and the increasing involvement of large commercial and investment banks (through their dealer shelves) in the sector. As HPA rates began to decrease in 2006, the spread between these two sectors expanded as market participants began to demand a greater risk premium for taking on the credit risk associated with alt-A borrowers.

While secondary market liquidity may wax and wane, in the long term we believe that the alt-A sector will remain viable as the market’s understanding of how different borrower and collateral characteristics determine prepayment and credit performance continues to improve. In addition, another source of loans for the sector may arise from disparities in the way which the GSEs and capital markets value convexity and credit. At the risk of providing readers with a blinding glimpse of the obvious, we believe that collateral characteristics in the sector will continue to evolve and the best opportunities will be available to investors who do their prepayment and credit homework.

Figure: 43. Spread between alt-A and Conforming Mortgage Rates: 1993-Current

Source: Loan Performance, Banc of America Securities

The Alt-B Sector

Another significant thread in the evolution of the alt-A sector has been the expansion of lender underwriting matrixes to provide mortgage credit to borrowers in the grey zone between subprime and alt-A. In particular, this trend in credit tiering has led to the creation of the Alternative-B (alt-B) sector. As with the alt-A sector, the alt-B sector defies easy categorization. At a broad level of generalization, if the alt-A sector represents the extension of expanded criteria (high LTVs, low or no documentation etc) underwriting to borrowers with high credit scores, the alt-B sector reflects a continuation of this trend to borrowers with lower credit scores.

IMPORTANT INFORMATION CONCERNING U.S. TRADING STRATEGISTS

Trading desk material is NOT a research report under U.S. law and is NOT a product of a fixed income research department of Banc of America Securities LLC, Bank of America, N.A. or any of their affiliates (collectively, “BofA”). Analysis and materials prepared by a trading desk are intended for Qualified Institutional Buyers under Rule 144A of the Securities Act of 1933 or equivalent sophisticated investors and market professionals only. Such analyses and materials are being provided to you without regard to your particular circumstances, and any decision to purchase or sell a security is made by you independently without reliance on us.

Any analysis or material that is produced by a trading desk has been prepared by a member of the trading desk who supports underwriting, sales and trading activities.

Trading desk material is provided for information purposes only and is not an offer or a solicitation for the purchase or sale of any financial instrument. Any decision to purchase or subscribe for securities in any offering must be based solely on existing public information on such security or the information in the prospectus or other offering document issued in connection with such offering, and not on this document.

Although information has been obtained from and is based on sources believed to be reliable, we do not guarantee its accuracy, and it may be incomplete or condensed. All opinions, projections and estimates constitute the judgment of the person providing the information as of the date communicated by such person and are subject to change without notice. Prices also are subject to change without notice.

With the exception of disclosure information regarding BofA, materials prepared by its trading desk analysts are based on publicly available information. Facts and ideas in trading desk materials have not been reviewed by and may not reflect information known to professionals in other business areas of BofA, including investment banking personnel.

Neither BofA nor any officer or employee of BofA accepts any liability whatsoever for any direct, indirect or consequential damages or losses arising from any use of this report or its contents.

To our U.K. clients: trading desk material has been produced by and for the primary benefit of a BofA trading desk. As such, we do not hold out any such research (as defined by U.K. law) as being impartial in relation to the activities of this trading desk.

IMPORTANT CONFLICTS DISCLOSURES

Investors should be aware that BofA engages or may engage in the following activities, which present conflicts of interest:

- The person distributing trading desk material may have previously provided any ideas and strategies discussed in it to BofA’s traders, who may already have acted on them.
- BofA does and seeks to do business with the companies referred to in trading desk materials. BofA and its officers, directors, partners and employees, including persons involved in the preparation or issuance of this report (subject to company policy), may from time to time maintain a long or short position in, or purchase or sell a position in, hold or act as market-makers or advisors, brokers or commercial and/or investment bankers in relation to the products discussed in trading desk materials or in securities (or related securities, financial products, options, warrants, rights or derivatives), of companies mentioned in trading desk materials or be represented on the board of such companies. For securities or products recommended by a member of a trading desk in which BofA is not a market maker, BofA usually provides bids and offers and may act as principal in connection with transactions involving such securities or products. BofA may engage in these transactions in a manner that is inconsistent with or contrary to any recommendations made in trading desk material.
- Members of a trading desk are compensated based on, among other things, the profitability of BofA’s underwriting, sales and trading activity in securities or products of the relevant asset class, its fixed income department and its overall profitability.
- The person who prepares trading desk material and his or her household members are not permitted to own the securities, products or financial instruments mentioned.
- BofA, through different trading desks or its fixed income research department, may have issued, and may in the future issue, other reports that are inconsistent with, and reach different conclusions from the information presented. Those reports reflect the different assumptions, views and analytical methods of the persons who prepared them and BofA is under no obligation to bring them to the attention of recipients of this communication.

This report is distributed in the U.S. by Banc of America Securities LLC, member NYSE, NASD and SIPC. This report is distributed in Europe by Banc of America Securities Limited, a wholly owned subsidiary of Bank of America NA. It is a member of the London Stock Exchange and is authorized and regulated by the Financial Services Authority.